Comparing Vision Transformers (ViT) and Hiera Transformers for Skin Cancer Classification

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***Abstract*—The research evaluates the performance of Vision Transformers together with Hiera Transformers for identifying skin cancer from dermatological images. Early skin cancer detection serves as a fundamental factor for achieving positive patient outcomes because this disease frequently appears among the most prevalent types of cancer. The strong classification capabilities of Vision Transformers come at the cost of extensive resource requirements. Hiera Transformers present a less demanding computing solution that retains promising results. This research implements HAM10000 dataset testing on the studied models through accuracy and training time and memory usage measurements to establish practical Ness for medical imaging applications.**

***Index Terms*—Skin cancer classification, vision transformers (ViT), hiera transformers, medical imaging, deep learning, HAM10000 dataset, transformer models, attention mechanism, computational efficiency, AI in healthcare.**

I. Introduction and Statement of the Problem

Data from the World Health Organization (2023) shows that skin cancer stands among the most common types of cancer worldwide as it continues to increase its occurrence since the previous several decades. The timely detection of skin cancer through accurate diagnosis leads to improved treatment effectiveness and simplification of medical interventions. Physicians performing dermatological analysis have traditionally used their manual skills and dermatoscope images as they train their algorithms by implementing Convolutional Neural Networks (CNNs). CNNs have enhanced diagnosis accuracy but the network cannot identify long-range dependencies nor understand entire image features (Dosovitskiy et al., 2020).

Vision Transformers (ViTs) brought forth a new image classification model architecture that uses self-attention mechanisms to extract contextual information from the entire image. The benchmark tasks show that ViTs outperform CNNs in delivering better performance results. The excessive resource usage of these methods creates obstacles for real-time medical applications together with low-resource environments (Dahmani, Tarhouni, Zidi, 2024).

The lightweight Hiera Transformers represent an updated version of transformers which cuts down computational expenses while maintaining accuracy performance (Hiera Transformer Research, 2023). The evaluation of Hiera Transformers for medical image classification applications remains limited especially within dermatology context. The research compares the performance of Hiera Transformers against ViTs when used for skin cancer diagnoses employing the HAM10000 dataset (Tschandl, Rosendahl, Kittler, 2018). Researchers aim to verify if Hiera Transformers function as an efficient computational substitute to ViTs while delivering equivalent diagnostic precision. The research targets to enhance AI-assisted dermatology through addressing this problem while establishing guidelines for selecting models for medical deployment.

II. Review of Literature

Machine learning has completely changed how we detect and classify images, opening the door to countless uses. Early on, methods like support vector machines (SVMs) and decision trees were common, but they needed humans to carefully pick and craft the features. Then came deep learning, which flipped the script. Convolutional Neural Networks (CNNs) took center stage, automatically figuring out complex patterns and features straight from the images themselves . CNNs took image classification and object detection to new heights, boosting both accuracy and speed. They've made a huge impact everywhere, from spotting issues in medical scans to helping self-driving cars navigate. The game changed again with Vision Transformers (ViTs). These treat chunks of an image like words in a sentence, letting them grasp the big picture and how everything connects. This is a game-changer for tricky situations where understanding the whole context is key [Ali et al., 2021].

Vision Transformers (ViTs) haven't just added to what deep learning can do—they've sparked fresh possibilities for working with images. ViTs are popping up in all sorts of image detection and classification jobs because they're great at seeing the whole picture and how everything in it connects. In healthcare, they're shining bright, especially for spotting diseases in X-rays and other scans, becoming a powerful ally in tackling illnesses like COVID-19 [ALKolifi ALEnezi, 2019] Recent advancements in hierarchical vision transformers have packed in specialized features to boost performance in tasks like classifying images. These additions improve accuracy and reduce computational demands, but they also make the models more complex, slowing them down compared to the simpler, original Vision Transformers (ViTs). To tackle this, researchers developed Hiera, a streamlined hierarchical vision transformer that's not only more accurate but also much faster during both training and real-world use [Azad et al., 2024]. When applied to skin cancer detection, Hiera could be a game-changer. By analyzing dermoscopic images—like those from the HAM10000 or ISIC datasets—Hiera's ability to quickly and accurately identify patterns in skin lesions could help spot melanoma or other skin cancers early. Its speed means it could process images in real-time, assisting dermatologists in making faster, more reliable diagnoses, especially in high-stakes cases where catching cancer early can save lives. Plus, its efficiency could make it practical for use in mobile apps or resource-limited clinics, bringing advanced diagnostics to more people [Dahmani et al., 2024].

III. Objectives of the Study

The primary objectives of this project include:

- This research examines skin cancer classification through Evaluation of Vision Transformers (ViTs) and Hiera Transformers implemented on the HAM10000 dataset where accuracy and precision and recall and F1-score metrics determine model comparisons.
- This study evaluates ViT and Hiera Transform models for computational efficiency by assessing training duration as well as memory demands and equipment requirements to select the most suitable model between efficiency and performance considerations.
- A practical usability evaluation of Hiera Transformers should be conducted for medical applications that operate with restricted computational power as well as the verification of diagnostic precision in such environments.

IV. Data Collection

TThe dataset used for this research is the HAM10000 dataset, which is publicly available on Harvard dataset . It contains 10,014 labelled dermoscopic images from seven different skin lesion categories. The dataset was sourced from multiple institutions and serves as a comprehensive collection for training deep learning models for skin cancer classification. The initial preprocessing steps were essential to optimize the data quality:

- **Resizing:** All images were resized to a uniform dimension of 224x224 pixels to ensure compatibility with the Vision Transformer (ViT) and Hiera Transformer models.
- **Normalization:** Pixel values were normalised to the range [0, 1] to increase model performance and convergence during training.
- **Augmentation:** Various augmentation techniques, including rotation, flipping were applied to create more diverse training data and reduce the potential for overfitting.
- **Data Splitting:** The dataset was divided into training, validation sets using a 80% and 20% split ratio to ensure balanced representation and model validation.

Data preprocessing steps standardize the modeling dataset which improved both training efficiency and evaluation capabilities resulting in better classification model accuracy and generalizability.

V. Exploratory Data Analysis (EDA) and Hypotheses for the Study

A. Exploratory Data Analysis

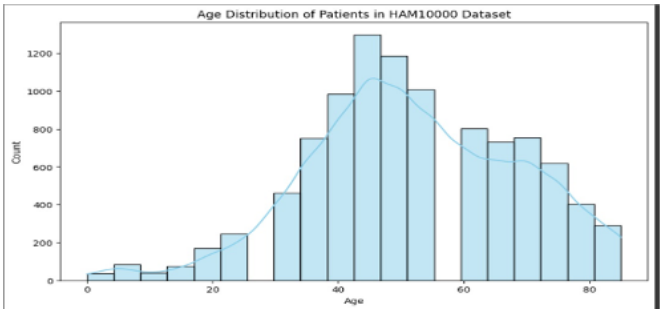

Fig. 1. Age Distribution of patients

The above histogram with a KDE shows most patients are between 35 and 65 years old. Fewer patients are in the very young and elderly age groups. The distribution is slightly right skewed, indicating that there are more middle-aged people.

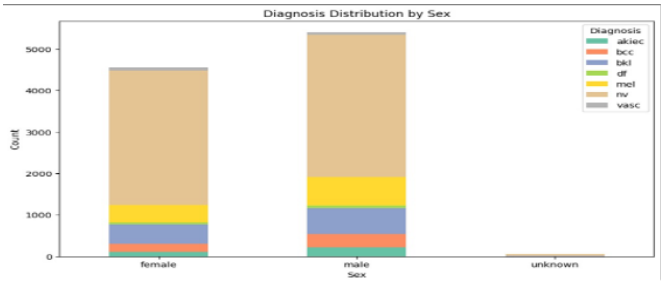

Fig. 2. Diagnosis Distribution by Sex

The above bar chart shows the distribution of skin diseases across sex category. The most frequent condition across all genders is nevus. male patients slightly outnumber females and show higher counts across most diagnosis categories

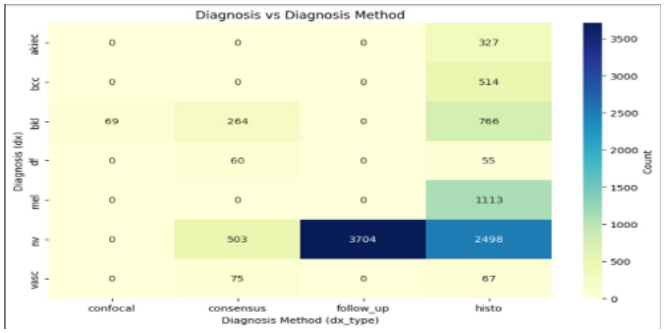

Fig. 3. Diagnosis vs Diagnosis Method

The above heatmap shows the relationship between diagnosis type and diagnosis method. The most common doagnosis nv(nevus) was predominantly determined by follow-up and histopathology.

B. Hypotheses

Based on the findings from the EDA, we formed the following hypotheses:

- **H0:**There is no significant difference in classification performance between Vision Transformers (ViTs) and Hiera Transformers.
- **H1:** Hiera Transformers are computationally more efficient while providing similar or better classification accuracy than Vision Transformers.

VI. Data Analytics

The complete data analytics workflow comprises several key stages:

A. Preprocessing Steps

- Image resizing, normalization, and augmentation were applied to prepare the dataset for model training.
- The data was split into training and validation,(80%, 20% respectively).

B. Training Models

- Vision Transformers (ViT-B/16) and Hiera Transformers were implemented using the AutoModelForImageClassification from the Hugging Face transformers library.
- Optimizers: AdamW with learning rate scheduling
- Early stopping was used to prevent overfitting during training.

C. Evaluation Metrics

- Accuracy: Proportion of correctly classified images.
- Precision, Recall, F1-Score: Used for a more detailed performance analysis.
- Memory usage, Computational Time, Accuracy: are used for model performance metrics

VII. Data Visualization and Results Report

A. Research Design Workflow

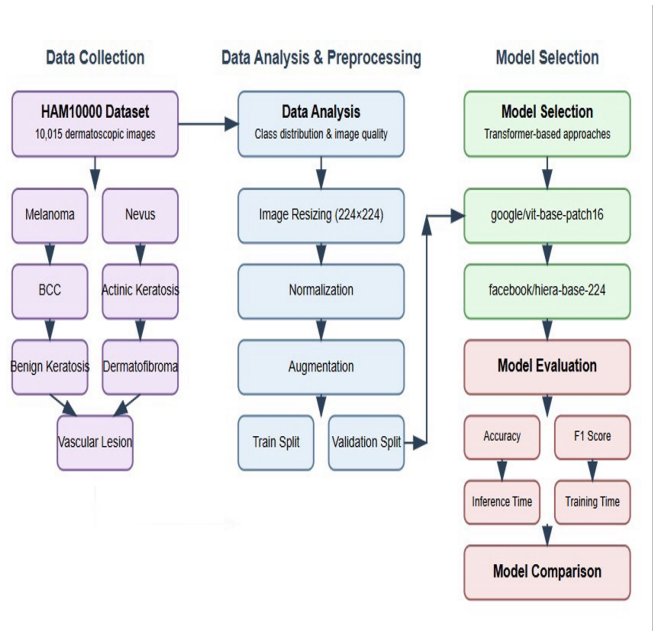

Fig. 4. Research Design Workflow

Workflow Explanation

The workflow design illustrates the data collection and preprocessing then evaluation processes for skin cancer classification using transformer models.

Workflow Overview

1. Data Collection

Gathered the HAM10000 dataset with 10,015 dermatoscopic images, categorized by skin conditions like Melanoma, Nevus, etc., for model training.

2. Data Preprocessing

Resize images to 224×224, normalize pixel values, and apply augmentation to increase dataset diversity, followed by splitting data into train and validation sets.

3. Model Selection

Selected the Google ViT and Facebook Hiera Transformer for their advanced image classification capabilities, leveraging transformer-based architecture for feature extraction.

4. Model Evaluation

Measured model performance using accuracy, F1 score, and training time to assess both accuracy and efficiency.

5. Model Comparison

Compared ViT and Hiera Transformer based on performance metrics to determine the best model for skin cancer classification.

B. Model Architecture

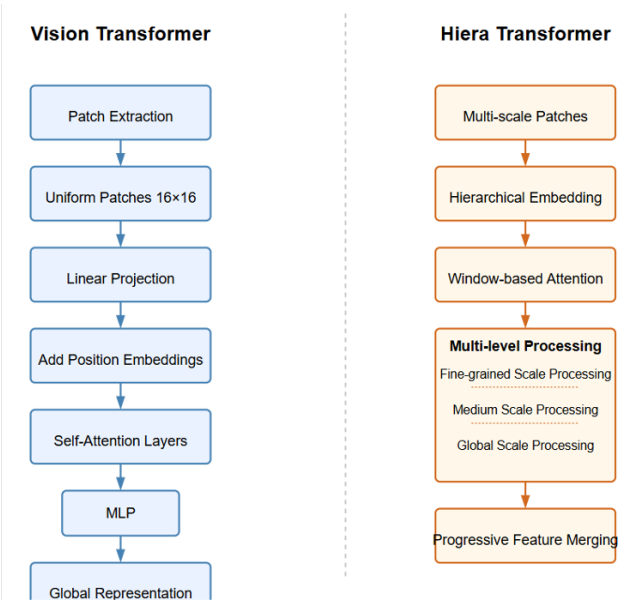

Fig. 5. Hiera & Vision Transformers Architecture

The above image demonstrates the architectural comparison between Hiera Transformer and Vision Transformer. This analysis highlights their separate methods of feature extraction and data processing. Efficient classification of skin cancer images occurs through advanced transformer-based techniques in all steps and models.

1. Vision Transformer

ViT breaks down the image into 16×16 patches and processes them sequentially to extract features.

- **Patch Extraction:** The image is divided into uniform patches (16×16).
- **Linear Projection:** These patches are then linearly projected to a higher-dimensional space.
- **Position Embeddings:** Position embeddings are added to the patches to retain spatial information.
- **Self-Attention Layers:** These layers allow the model to focus on important parts of the image by assigning attention weights to different regions of the image.
- **MLP (Multilayer Perceptron):** A fully connected layer to aggregate the output from attention mechanisms.
- **Global Representation:** The final representation is global, capturing the entire image's features.

2. Hiera Transformer

Hiera Transformer uses a more advanced approach by employing multi-scale patch processing and hierarchical feature extraction.

- **Multi-scale Patches:** Unlike ViT, Hiera Transformer extracts patches at multiple scales to capture both fine-grained and coarse features.
- **Hierarchical Embedding:** This step provides a hierarchical representation of image features, improving the model's ability to understand complex structures.
- **Window-based Attention:** Attention is computed within local windows, which helps the model focus on important regions of the image.
- **Multi-level Processing:** The model processes features at three levels: fine-grained, medium-scale, and global-scale, ensuring it captures both local and global dependencies in the image.
- **Progressive Feature Merging:** Features from various scales are progressively merged, ensuring a comprehensive understanding of the image at multiple levels.

C. Evaluation Metrics

For vision transformer:

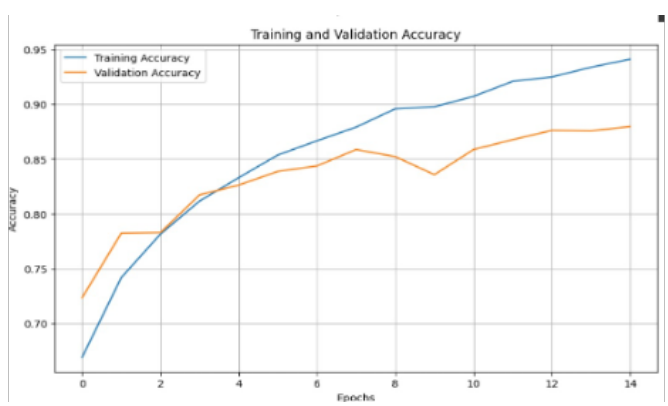

Fig. 6. Training and Validation Accuracy(ViT)

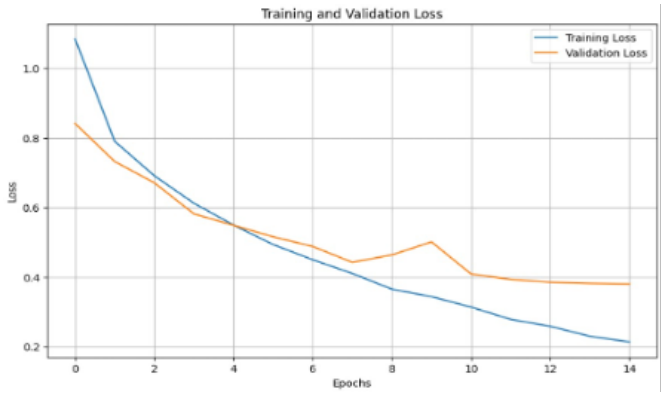

Fig. 7. Training and Validation Loss(ViT)

The Vision Transformer (ViT) demonstrates continuous accuracy and loss improvement across 15 training epochs. The model demonstrates rapid training accuracy elevation as validation accuracy matches at the same time the model learns successfully and applies learned patterns to new data inputs.

For Hiera transformer:

Fig. 8. Training and Validation Accuracy(Hiera)

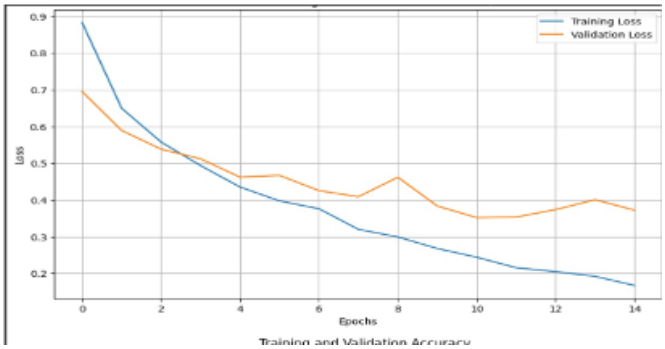

Fig. 9. Training and Validation Loss(Hiera)

The Hiera Transformer visualization reveals both progressive training accuracy improvement along with somewhat inconsistent validation accuracy behavior. Good learning performance emerges from the training data as both training loss declines steadily alongside the validation loss which shows slower reduction.

D. Model Performance Metrics

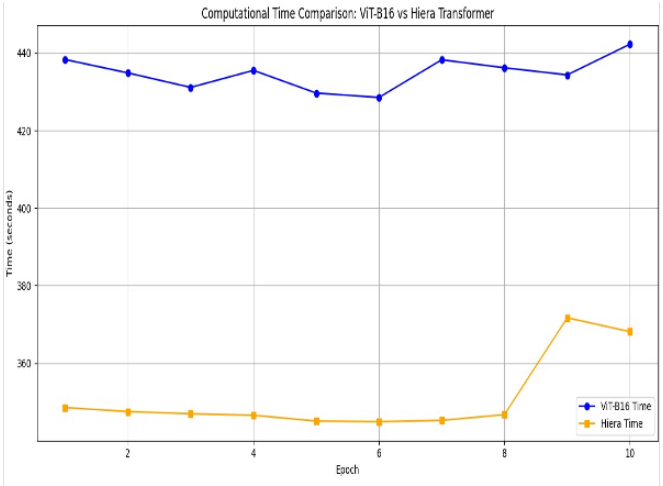

Fig. 10. Computational Time Comparison

The above graph clearly shows that Hiera Transformer took much less time to train per epoch compared to ViT-B/16. Even though Hiera's training time increased slightly after the 8th epoch, it was still way faster overall, making it a better option for quicker model training.

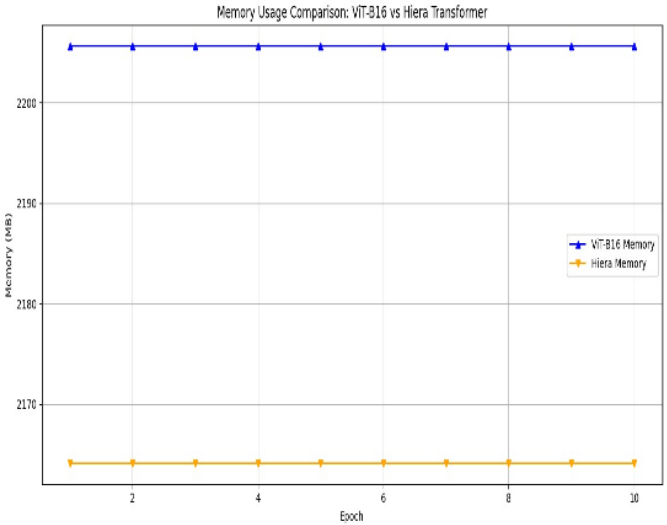

Fig. 11. Memory Usage Comparison

The above graph shows that ViT-B/16 used much more memory than the Hiera Transformer during all 10 training rounds. Hiera Transformer is more memory-friendly and a better option when there is limited memory available.

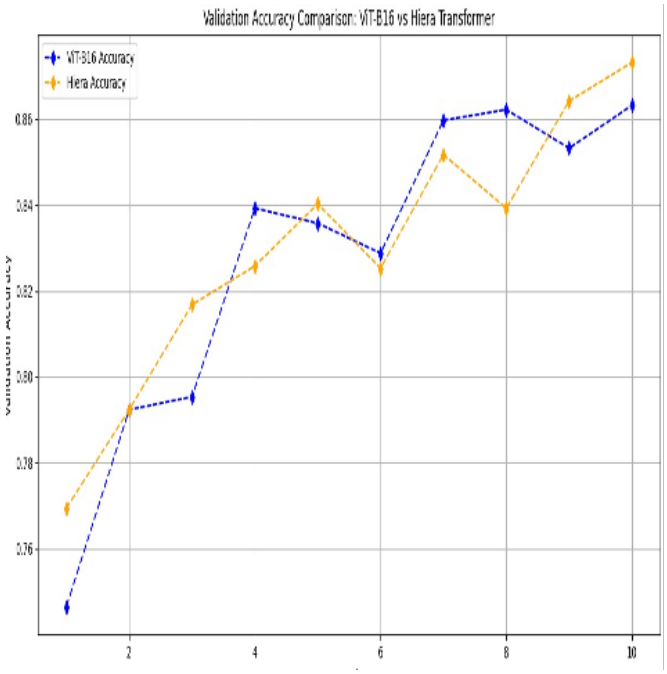

Fig. 12. Validation Accuracy Comparison

From the graph, we can see that both models improved over time, but Hiera Transformer slightly outperformed ViT-B/16 in the later stages. This suggests that Hiera not only uses less memory but can also achieve better or similar accuracy, making it a strong choice overall.

VIII. Conclusion

This research explored the performance comparison between Vision Transformers (ViTs) and Hiera Transformers when applied to classify skin cancer specimens from the HAM10000 dataset. The computational cost and memory needs put ViTs at a disadvantage for achieving their high accuracy rate. The trial showed that Hiera Transformers delivered equivalent or better results during classification tasks while consuming less training time and system resources. An analysis shows that Hiera Transformers present a time-efficient approach that maintains precise diagnostic capabilities. Research evidence demonstrates transformer-based models' capacity to enhance diagnostic performance while reducing healthcare process times. Research indicates that medical facilities serving under-resourced areas should implement Hiera Transformers systems. The findings support AI-assisted dermatology development by giving essential information for clinical implementations. These models demonstrate potential to provide essential benefits which would assist dermatologists when detecting skin cancer at its early stages.

IX. Future Scope

Further research needs to examine how combining Vision Transformers (ViTs) with Hiera Transformers will allow systems to benefit from each architecture's core strengths. The model's diagnostic accuracy together with its ability to generalize would benefit from incorporating both patient demographic information and medical chronicles with visual data. Real-time clinical data and large-scale dataset assessment will demonstrate scalability potential which enables evaluation of these models for future deployment in healthcare facilities. AI-assisted skin cancer detection needs optimization for deployment on mobile phones and handheld diagnostic devices to provide access in resource-limited settings.

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